

The law of logarithms and worked examples

When the rules of indices are expressed in logarithm form, we produce the very important laws of logarithms. These solve many complex

laws enable us to equations.

The Laws of Logarithms

$$1. \log_a(xy) = \log_a(x) + \log_a(y)$$

$$2. \log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$$

$$3. \log_a(x^n) = n\log_a(x)$$

$$4. \log_a(a) = 1$$

$$5. \log_a(1) = 0$$

$$6. \log_a(x) = \frac{\log_b(x)}{\log_b(a)}$$

Using your calculator, verify, each of the following:

$$1. \log_{10}(2) + \log_{10}(4) = \log_{10}(8) = 0.9031$$

$$2. \log_{10}(9) - \log_{10}(3) = \log_{10}\left(\frac{9}{3}\right) = 0.4771$$

$$3. \log_{10}(4^3) = 3\log_{10}(4) = 1.8062$$

$$4. \log_{10}(10) = 1$$

$$5. \log_{10}(1) = 0$$

The rules of logs apply to any base, however, the two most widely-used bases in logs are the base 10 and base e (≈ 2.718)

Base ten logs e.g. **$\log_{10}(100)$** , are used for calculation purposes and are referred to as common logs.

Base e (≈ 2.718) e.g. **$\log_e 1(00)$** , is used when dealing with naturally-occurring events, e.g. earthquakes, growth of colonies and hence are called natural logs and are written **$\log_e(x) = \ln(x)$** .

So, lets work through some examples applying some of the laws of logarithms.

Worked example 1:

Without using a calculator, simplify the following:

$$2 \log_{10}(3) + \log_{10}(16) - 2 \log_{10} \left(\frac{6}{5} \right)$$

This law was used here:
 $\log_a x^n = n \log_a x$

$$\Rightarrow \log_{10}(3^2) + \log_{10}(16) - \log_{10} \left(\frac{6}{5} \right)^2$$

This law was used here:
 $\log_a x^n = n \log_a x$

$$\Rightarrow \log_{10}(9 \times 16) - \log_{10} \left(\frac{36}{25} \right)$$

This law was used here:
 $\log_a xy = \log_a x + \log_a y$

$$\Rightarrow \log_{10} \left(\frac{9 \times 16}{\frac{36}{25}} \right)$$

This law was used here:
 $\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$

$$\Rightarrow \log_{10}(100)$$

So let

$$10^x = 100$$

$$\Rightarrow 10^x = 10^2$$

$$\Rightarrow x = 2$$

Therefore:

$$2 \log_{10}(3) + \log_{10}(16) - 2 \log_{10} \left(\frac{6}{5} \right) = \log_{10}(100) = 2$$

Worked example 2

Without using a calculator, simplify the following:

$$\log_2(64) + \log_3(135) - \log_3(5)$$

- $\log_2 64 + \log_3 \frac{135}{5}$
- $\log_2 64 + \log_3 27$

This law was used here:
 $\log_a \left(\frac{x}{y} \right) = \log_a(x) - \log_a(y)$

Since the bases are different, these logs cannot be added!

So, let

- $\log_2 64 = x$
- $2^x = 64$
- $2^x = 2^6$
- $x = 6$

Also, let,

- $\log_3 27 = y$
- $3^y = 27$
- $3^y = 3^3$
- $y = 3$

Therefore:

$$\log_2 64 + \log_3 135 - \log_3 5 = 6 + 3 = 9$$

Worked example 3

Without using a calculator, simplify the following:

$$\log_c(16) \times \log_2(c)$$

Note both terms have different bases!

We need to change base 2 to base c (or visa versa!)

$$\begin{aligned} \text{➤ } \log_2(c) &= \frac{\log_c(c)}{\log_c(2)} \\ \text{➤ } \log_2(c) &= \frac{1}{\log_c 2} \end{aligned}$$

Taking just one term, we use this law
 $\log_a x = \frac{\log_b x}{\log_b a}$

Insert this back in the original statement

$$\text{➤ } \log_c(16) \times \frac{1}{\log_c(2)} = \frac{\log_c(16)}{\log_c(2)}$$

But,

$$\text{➤ } \frac{\log_c(16)}{\log_c(2)} = \log_2(16) = \log_2(2^4) = 4$$

Using this law again!
 $\log_a x = \frac{\log_b x}{\log_b a}$

Therefore:

$$\log_c(16) \times \log_2(c) = 4$$

Worked example 4

Solve for x

$$4^x = 9$$

This law was used here:
 $\log_a x^n = n \log_a x$

$$\Leftrightarrow \log(4^x) = \log(9)$$

$$\Leftrightarrow x \log(4) = \log 9$$

$$\Leftrightarrow x = \frac{\log 9}{\log 4}$$

$$\Leftrightarrow x = \log_4(9)$$